Health Predictor Model

Meeting note 2026-10-30

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Project Overview

The Health Predictor Model project is a student-led exercise in building a cardiovascular disease prediction system. Members will work through the machine learning process from dataset selection to model evaluation and deployment. The finished system is intended as a learning project, not a clinical diagnostic tool.

Scope and Project Goals

The project covers:

- dataset selection,
- data cleaning
- exploratory data analysis (EDA)
- Feature engineering
- model training
- model interpretation
- API
- Documentation
- Final presentation.

This will not be (Outside of Project Scope):

- Clinical diagnosis
- Medical advice or treatment recommendations
- Handling protected health information
- large-scale cloud infrastructure

Software and Tools

- Python; Pandas and NumPy for data preparation and numerical work.
- Matplotlib and Seaborn for charts and exploratory analysis.
- Scikit-learn for baseline models and evaluation; XGBoost and LightGBM for more advanced models.
- FastAPI for the prediction API; Docker for a reproducible deployment.
- GitHub for source control, issues, pull requests, and task tracking.

Team Roles and Expected Work

Members will discuss their interests and experience before roles are assigned. Roles describe the main area of responsibility; members may help across workstreams as needed.

Machine Learning Workflow

1. Data acquisition
2. Cleaning and validation
3. Exploratory data analysis
4. Feature engineering
5. Baseline models
6. Advanced models
7. Evaluation
8. Experiment tracking
9. Deployment

Updated Project Schedule

Dates (2026) Planned work	
Oct 27 – Nov 2	Onboarding
Nov 2 - Nov 6	Environment Setting, VScode, Git/GitHub
Nov 3 – Nov 16	Data Selection, Data cleaning, validation, and exploratory analysis
Nov 17 – Nov 23	Feature engineering
Nov 24 – Nov 30	Baseline model training
Dec 1 – Dec 7	Advanced models and tuning
Dec 8 – Dec 14	Model interpretation and final evaluation
Dec 15 – Dec 20	Deployment and documentation

Team Communication

- Use Discord as the main team communication channel.
- Share meeting notes and agreed action items after each meeting.

Weekly Workload Updates (#Bullet Point format)

Name	Delieverables (What have you worked on last week?)	Projected Work this Week (What will you work on this week?)
Destiny	- N/A	-
Elliot	- N/A	-
Moksha	- N/A	-
Hendrick	- N/A	-
Shahar	- N/A	-

Previous Meeting Notes Oct. 30th

- Introductions and sneak-peek in AI concept and AI workflow
- Workload and communication expectations
- Project aims and success criteria
- Explanation on how we will update each other
- GitHub Issue assignment was distributed
- Last minute changes to Discord and Member arrangements (Yoon)

Final Deliverables by Nov 6th Meeting

- Members finished setting up **VScode** and are familiar with **Git/GitHub**
- Members should be able to explain Git issues and how work is distributed / closed
- Issue #1 and Milestone 1 is reached via GitHub